Elevator Trim Position Transmitter – General data

1. General

A potentiometer is mounted on the port side elevator at approximately station WL 54.00 and sends a signal of the elevator trim tab position through the DAU and onto the Flight Data Recorder (if installed) and the elevator trim tab position is indicated on the system screen.

2. Removal/Installation

Equipment and Materials:

- Corrosion preventative compound MIL–C–16173 Grade 2 or equivalent

A. Remove the Elevator Trim Indicator Potentiometer

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

3) Disconnect the electrical connection at the terminal block.

4) Remove the cotter pin, (2) washers and clevis pin from the push rod end and link halves.

5) Remove the cotter pin, (2) washers and the clevis pin from the link halves to the lever assembly.

6) Loosen the nut, and screw that hold the lever assembly to the potentiometer shaft mounted in mounting bracket. Remove lever assembly.

7) Remove two bolts holding the potentiometer mounting bracket to the bracket assembly.

8) Remove the safety wire and the (3) screws and clamps which secure the potentiometer onto the mounting bracket.

9) Remove potentiometer.

B. Install the Elevator Trim Indicator Potentiometer

Note Coat all bolt shanks with corrosion preventative compound MIL–C–16173 Grade 2 or equivalent before installation.

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Ensure the FLIGHT CONTROLS INDICATION (A14) circuit breaker is pulled and fitted with safety clip.

3) Position the potentiometer on to the mounting bracket. Secure with the (3) clamps and screws. Safety wire screws.

4) Position the mounting bracket on the bracket assembly and secure in place with bolt and washers in two places.

5) Locate the lever assembly on potentiometer shaft leaving a gap of 0.02 to 0.05 of an inch between lever assembly and the potentiometer housing to minimize the pushrod assembly misalignment.

6) Secure with a washer nut and screw.

7) Install the clevis pin, (2) washers cotter pin from the end of the lever assembly to the link halves.

8) Install the clevis pin, (2) washers cotter pin from the end of the link halves to the end of the push rod assembly.

9) Ensure that the mechanism can move freely through the full range of motion.

10) Adjust the elevator trim tab indicator potentiometer. (Refer to Para 3.A).

11) Reconnect the electrical connection at the terminal block.

12) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

13) Do a function test per Para 3.B.

3. Adjustment/Test

A. Adjustment Elevator Trim Tab Indicator Potentiometer

Equipment and Material:

- DC Voltmeter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Measure the Ground Power Voltage (V_{GP}) at the external power relay (or other suitable location).

3) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit with safety clip.

4) Move elevator to neutral position and lock in place. Set elevator trim tab to the neutral position.

5) Disconnect terminal 2 of the potentiometer electrical connection at the terminal block and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION **ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.**

6) Loosen screw and nut on lever at the potentiometer shaft enough that the shaft turns freely.

7) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker.

8) Rotate potentiometer shaft until the measured DC Voltage is precisely 50% of total aircraft voltage V_{GP} (approximately 14 volts).

9) Pull FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit with a safety clip.

10) Disconnect voltmeter from potentiometer. Reconnect pushrod and lever assembly to potentiometer shaft.

Note Do not rotate potentiometer shaft during this process.

11) Reconnect the potentiometer electrical connections at the terminal block.

12) Remove safety clip and reset FLIGHT CONTROLS INDICATION (A14) circuit breaker.

13) Remove elevator control lock.

14) Do a function test (refer to Para 3.B).

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.

15) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Function Test Elevator Trim Tab Indicator Potentiometer

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Wait several minutes for the aircraft electrical and avionics systems to initialize.

3) Deflect elevator trim to the full up position and note the Pitch Trim Display Indication correlates to approximately 20 degrees and does not become invalid (indicated by an amber 'X').

4) Deflect elevator trim to the neutral position and note the Pitch Trim Display Indication correlates to approximately 0 degrees.

5) Deflect elevator trim to the full down position and note the Pitch Trim Display Indication correlates to approximately 25 degrees and does not become invalid (indicated by an amber 'X').

6) If the Pitch Trim Display does not match the elevator trim surface position, repeat adjustment steps in Para 3.A.

7) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

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